Decomposing Theory of Mind: How Emotional Processing Mediates ToM Abilities in LLMs

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Abstract

Recent work shows activation steering substantially improves language models' Theory of Mind (ToM) (Bortoletto et al. 2024), yet *what* changes remains unclear. We decompose ToM by comparing steered versus baseline LLMs' activations using linear probes trained on 45 cognitive actions. Applying Contrastive Activation Addition (CAA) to 1,000 BigToM forward belief scenarios, we find successful ToM enhancement (32.5% to 46.7% accuracy) is mediated by activations processing emotional content*emotion_perception* (+2.23), *emotion_valuing* (+2.20), *noticing* (+2.08), while suppressing analytical processes—*questioning* (-0.78), *convergent_thinking* (-1.59). This reveals that successful ToM abilities in LLMs are mediated by emotional understanding, not analytical reasoning.

Introduction

Theory of Mind (ToM)—the capacity to attribute mental states to oneself and others—represents a critical capability for AI systems engaged in social reasoning, alignment, and human collaboration. Recent benchmarking work by Bortoletto et al. (2024) has demonstrated that activation steering techniques, particularly Contrastive Activation Addition (CAA), can substantially improve language models' ToM performance, often achieving accuracy improvements exceeding 10 percentage points. This raises a fundamental question: *what cognitive processes change* when models successfully engage in perspective-taking?

Traditional evaluations treat ToM as a monolithic capability measured through binary accuracy on belief attribution tasks. However, cognitive science views ToM as comprising multiple interacting processes: perceptual encoding, emotional inference, hypothesis generation etc. If steering vectors modulate these components differently, comparing steered versus baseline activation patterns could reveal which processes are *essential* for successful perspective-taking—effectively decomposing ToM into its constituent building blocks.

We introduce a mechanistic decomposition approach combining linear classifer probes trained on 45 cognitive actions with CAA steering vectors. By measuring which cognitive processes are systematically upregulated or downregulated when ToM performance improves, we can identify the fundamental components underlying successful perspective-taking.

Contributions: (1) A mechanistic decomposition methodology using cognitive action probes to analyze steering effects; (2) Evaluation on 1,000 BigToM forward belief scenarios showing 14.2 percentage point accuracy improvement (32.5% to 46.7%); (3) Discovery that successful ToM systematically upregulates emotional/creative processes while downregulating analytical processes; (4) Findings suggesting that emotional understanding serves as the fundamental building block of perspective-taking in language models.

Methodology

To investigate how steering vectors modulate cognitive processes during ToM tasks, we developed a multi-stage pipeline integrating probe training, activation steering, and comparative analysis. We began by defining 45 cognitive actions across four categories: *Metacognitive*, *Analytical*, *Creative*, and *Emotional* (see appendix). For each action, we generated 31,500 synthetic training examples (700 per action) using Gemma-3-4B. First-person narratives (2-4 sentences) varied across 20 everyday contexts, producing balanced datasets with realistic language patterns.

Activations were extracted from layers 0-30 of Gemma-3-4B using nnsight, loosely following the methodology from (Chen et al. 2024). To ensure consistent probe training, inputs were augmented (at extraction and during inference) with the suffix "The cognitive action being demonstrated here is" for consistent final-token extraction. We trained 45 binary linear probes using one-vs-rest classification with AdamW optimization, cosine annealing, and early stopping based on AUC-ROC (Alain and Bengio 2016).

For activation steering, we trained CAA vectors (Rimsky et al. 2023; Zou et al. 2023) on 752 contrastive triplets from BigToM's *forward_belief* scenarios (50/50 false-true split). Each triplet contains story context, belief question, and positive (correct ToM) versus negative (incorrect ToM) completions, capturing representational differences between accurate and inaccurate perspective-taking. Vectors were trained across layers 14-30 using PCA-centered activation differences.

We evaluated 1,000 forward belief scenarios from BigToM (*forward_belief_false*), comparing baseline versus

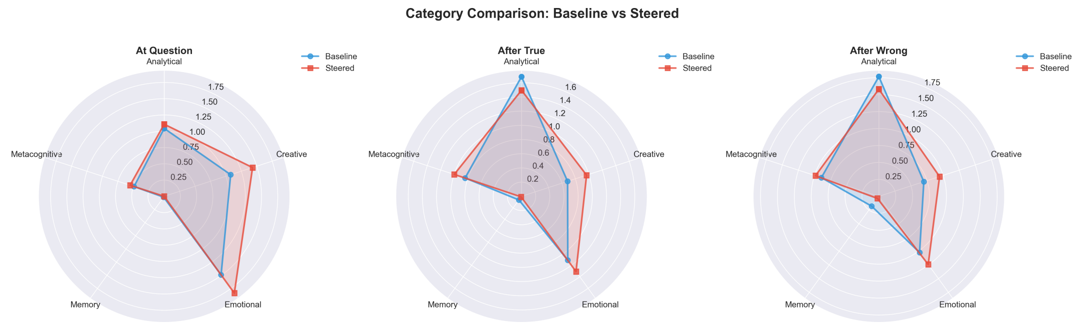

Figure 1: Radar chart comparing baseline versus steered cognitive action activation patterns across categories. The steered condition (red) compared to baseline (blue)

steered conditions. Answers were evaluated by computing p(correct) vs p(incorrect) from model logits. Cognitive action activations were captured at three timepoints: (1) at question, (2) after true answer, (3) after wrong answer. For each action, we computed layer count (layers 10-20 where probe confidence indicated presence) and analyzed baseline-steered differences to identify which cognitive processes characterize successful ToM improvements.

Results

We first validated our probe methodology, finding that binary probes achieved 0.78 average AUC-ROC and 0.68 F1 across 45 actions. Mid-layer performance (layers 5-24) outperformed early/late layers, informing our layer 10-20 analysis window. Having established probe validity, we evaluated CAA steering on 1,000 forward belief scenarios (false condition), observing accuracy improvements from 32.5% to 46.7% (14.2pp gain), shifting 217 examples from incorrect to correct predictions.

To understand what drives this improvement, we analyzed cognitive action activation patterns in baseline versus steered conditions. The most striking finding was robust upregulation of emotional and generative processes. Emotional processes showed strong increases: *emotion_perception* (mean Δ=+1.73), *emotion_valuing* (Δ=+0.85), and *emotion_understanding* (Δ=+0.77). Additionally, *Hypothesis_generation* (Δ=+1.63) remained strongly elevated across all timepoints, indicating active belief formation and explanation generation.

In contrast, analytical processes decreased: *questioning* (Δ=-1.24), *convergent_thinking* (Δ=-1.13), and *understanding* (Δ=-0.77), suggesting successful ToM suppresses deliberate analytical interrogation. Category aggregation reinforces this pattern—creative processes (+0.35, +0.28, +0.24) and emotional processes (+0.35, +0.20, +0.22) consistently increase across timepoints, while analytical processes show decreases (+0.06, -0.19, -0.19).

Discussion

This pattern of upregulated emotional and generative processes and downregulated analytical processes challenges assumptions that LLM social reasoning relies on deliberate chain-of-thought mechanisms. Rather, successful perspective-taking appears to operate through activating representations responsible for processing emotional contexts.

We make no claims that these findings generalize to human cognition, nor does our methodology validate such comparisons. However, neuroscience research shows that affective and cognitive ToM share neural mechanisms in humans: Corradi-Dell'Acqua, Hofstetter, and Vuilleumier (2014) demonstrate that "patterns in TPJ (Temporoparietal Junction) and MTG (Middle Temporal Gyrus) reflect the same neuronal activity, equally recruited in these two independent conditions." During language modeling, networks may learn shared representations linking perspective-taking with emotional context processing, mirroring the compressed structure of human social cognition embedded in linguistic data. Whether this constitutes genuine emulation of cognitive architecture or emergent convergence on functionally equivalent representations remains an open question.

Several limitations warrant consideration. Our probe training relies on synthetic data from a single small model (Gemma-3-4B). Bortoletto et al. (2024) already demonstrated how steering vectors behave and generalize across different BigToM conditions, which was not the focus of our work. Future work should validate these cognitive decomposition findings with multiple, bigger models.

Conclusion

Our work extends Bortoletto et al. (2024) by applying cognitive action probes to activation steered LLMs, we move beyond evaluating *whether* steering improves ToM to understanding *how* it works. Our analysis of 1,000 forward belief scenarios reveals systematic modulation of cognitive processes: steering amplifies generative hypothesis formation and emotional inference while suppressing analytical interrogation. This decomposition approach, combining interpretability tools with targeted interventions, offers a principled methodology for understanding complex AI capabilities. The findings challenge assumptions about deliberative social reasoning in LLMs and open new directions for mechanistic analysis of perspective-taking and other high-level cognitive abilities.

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Appendix

Complete Cognitive Action Taxonomy

The 45 cognitive actions used for probe training were derived from established cognitive psychology and emotion research frameworks, then organized into five functional categories. Actions were systematically sampled from Bloom's Taxonomy (Anderson et al. 2001), Guilford's Structure of Intellect (Guilford 1967), Metacognitive Processes Framework (Flavell 1979), Krathwohl's Affective Domain (Krathwohl, Bloom, and Masia 1964), Gross's Process Model of Emotion Regulation (Gross 1998), and the Mayer-Salovey Emotional Intelligence Model (Mayer and Salovey 1997).

Metacognitive (7 actions):

- *reconsidering*: reconsidering a belief or decision
- *updating_beliefs*: updating mental models or beliefs
- *suspending_judgment*: suspending judgment and staying with uncertainty
- *meta_awareness*: reflecting on one's own thinking process
- *metacognitive_monitoring*: tracking one's own comprehension
- *metacognitive_regulation*: adjusting thinking strategies
- *self_questioning*: interrogating one's own understanding

Analytical (16 actions):

- *noticing*: noticing a pattern, feeling, or dynamic
- *pattern_recognition*: recognizing recurring patterns across situations
- *zooming_out*: zooming out for broader context
- *zooming_in*: zooming in on specific details
- *questioning*: questioning an assumption or belief
- *abstracting*: abstracting from specifics to general patterns
- *concretizing*: making abstract concepts concrete and specific
- *connecting*: connecting disparate ideas or experiences
- *distinguishing*: distinguishing between previously conflated concepts
- *perspective_taking*: taking another's perspective or temporal view
- *convergent_thinking*: finding the single best solution
- *understanding*: interpreting and explaining meaning
- *applying*: using knowledge in new situations
- *analyzing*: breaking down into components
- *evaluating*: making judgments about value or effectiveness
- *cognition_awareness*: becoming aware and comprehending

Creative (6 actions):

- *creating*: generating new ideas or solutions
- *divergent_thinking*: generating multiple creative solutions
- *hypothesis_generation*: generating possible explanations
- *counterfactual_reasoning*: engaging in 'what if' thinking
- *analogical_thinking*: drawing analogies between domains
- *reframing*: reframing a situation or perspective

Emotional (15 actions):

- *emotional_reappraisal*: reinterpreting emotional meaning

- *emotion_receiving*: becoming aware of emotions
- *emotion_responding*: actively engaging with emotions
- *emotion_valuing*: attaching worth to emotional experiences
- *emotion_organizing*: integrating conflicting emotions
- *emotion_characterizing*: aligning emotions with core values
- *situation_selection*: choosing emotional contexts deliberately
- *situation_modification*: changing circumstances to regulate emotion
- *attentional_deployment*: directing attention for emotional regulation
- *response_modulation*: modifying emotional expression
- *emotion_perception*: identifying emotions in self/others
- *emotion_facilitation*: using emotions to enhance thinking
- *emotion_understanding*: comprehending emotional complexity
- *emotion_management*: regulating emotions in self/others
- *accepting*: accepting and letting go of control

Memory (1 action):

- *remembering*: recalling relevant information or experiences

CAA Training Data Example

The CAA steering vectors were trained on contrastive pairs distinguishing correct from incorrect belief attribution. Each training triplet consists of a story with observational information, a question about the protagonist's belief, and two completions representing accurate vs. inaccurate Theory of Mind reasoning.

False Belief Example:

Story: "Noor is working as a barista at a busy coffee shop. Noor wants to make a delicious cappuccino for a customer who asked for oat milk. Noor grabs a milk pitcher and fills it with oat milk. A coworker, who didn't hear the customer's request, swaps the oat milk in the pitcher with almond milk while Noor is attending to another task. Noor does not see her coworker swapping the milk."

Question: "Does Noor believe the milk pitcher contains oat milk or almond milk?"

Positive (correct ToM): "Noor believes the milk pitcher contains oat milk."

Negative (incorrect ToM): "Noor believes the milk pitcher contains almond milk."

Evaluation Methodology: Answer Ranking by Probability

Following the BigToM evaluation protocol, we used answer ranking by probability rather than free-form text generation. Each question was formatted as a multiple-choice task with randomized answer positions:

Prompt Format:

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Story: [story text]

Question: [question text]
Choose one of the following:
a) [answer 1]
b) [answer 2]

Please answer with the letter of your choice (a o
Answer:
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Evaluation Process: For each question, we:

1. Randomized the positions of true and wrong answers between options a) and b)
2. Calculated $p(\text{letter} = \text{`a'})$ and $p(\text{letter} = \text{`b'})$ from model logits at the final token position
3. Selected the answer corresponding to the letter with higher probability
4. Determined correctness by comparing the selected answer to the ground truth

This probability-based ranking approach eliminates confounds from text generation artifacts and provides a more reliable measure of the model's belief attribution capabilities. Answer position randomization ensures that the model cannot exploit systematic biases in option ordering.

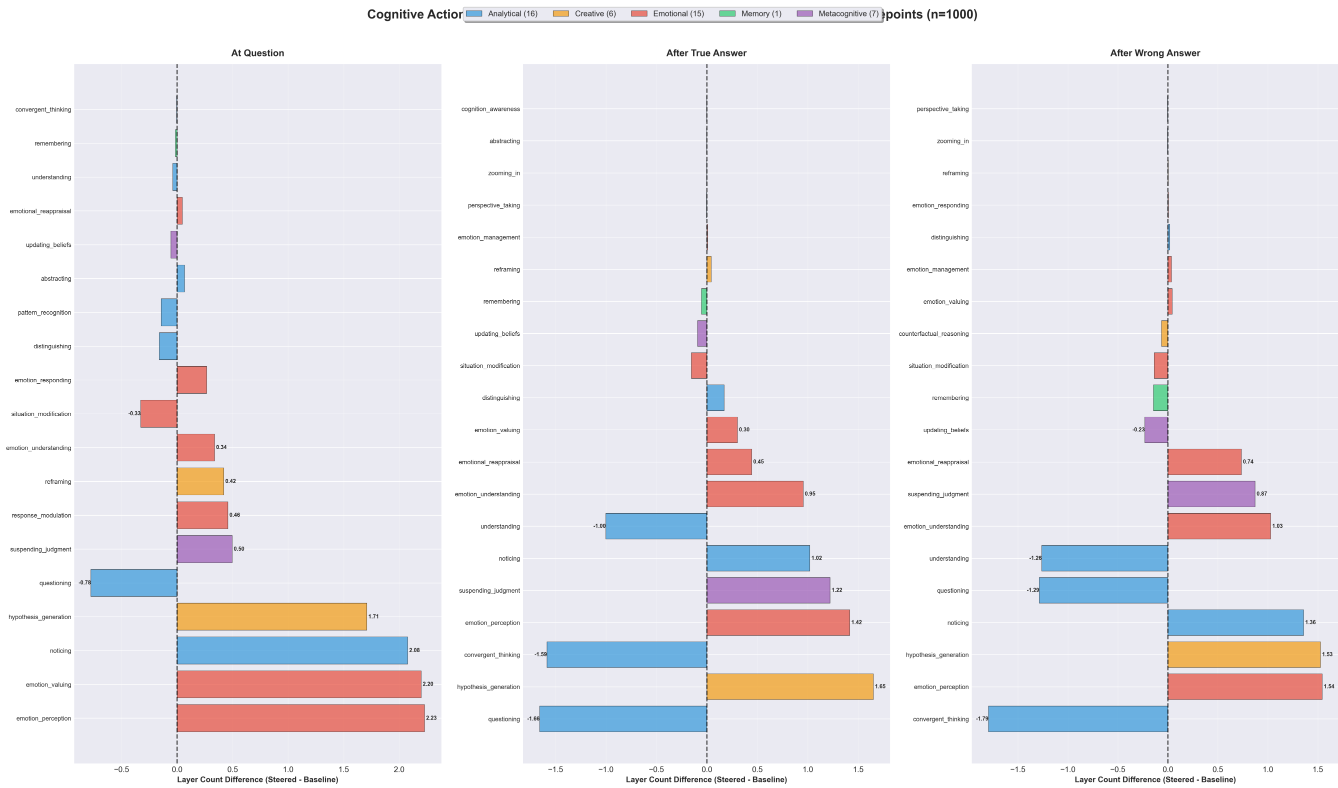

Figure 2: Steering effects across all cognitive actions and timepoints (n=1000). Left panels show individual action changes at three timepoints: at question (before answer), after true answer, and after wrong answer. Bars indicate mean layer count difference (steered - baseline) with positive values (right) showing increases and negative values (left) showing decreases. Emotional actions (*emotion_perception*, *emotion_valuing*, *noticing*) consistently increase across timepoints, while analytical actions (*questioning*, *convergent_thinking*, *understanding*) consistently decrease, revealing the emotional foundation of successful ToM.

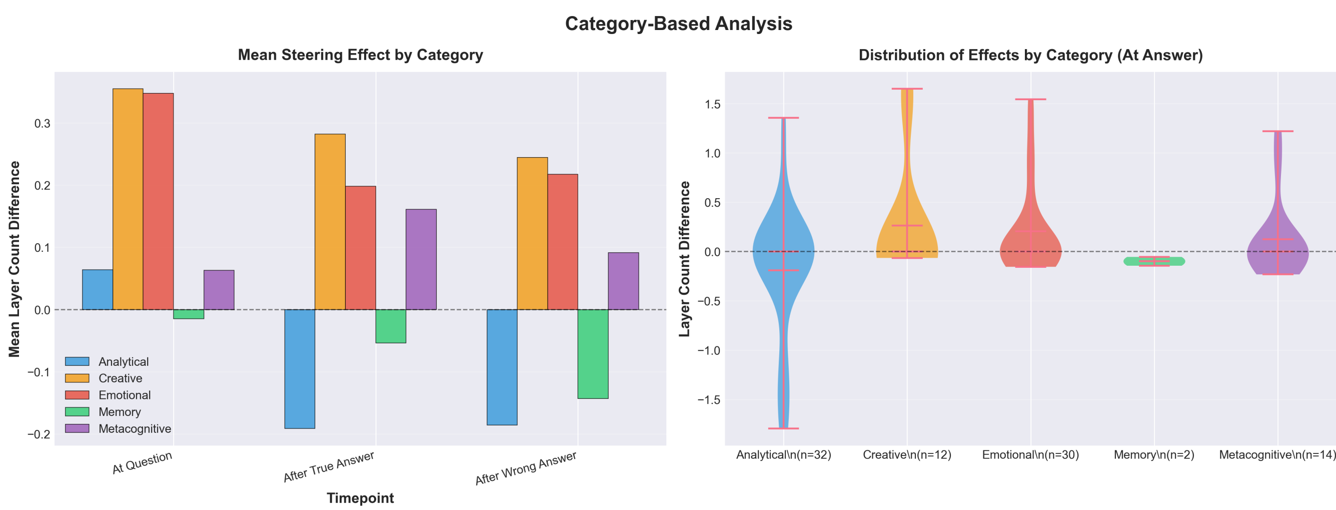

Figure 3: Category-level analysis of steering effects. Negative points represent more activation at baseline while positive represent more activation on steered. Left: mean steering effect by cognitive action category. Right: distribution of effects at answer timepoint.

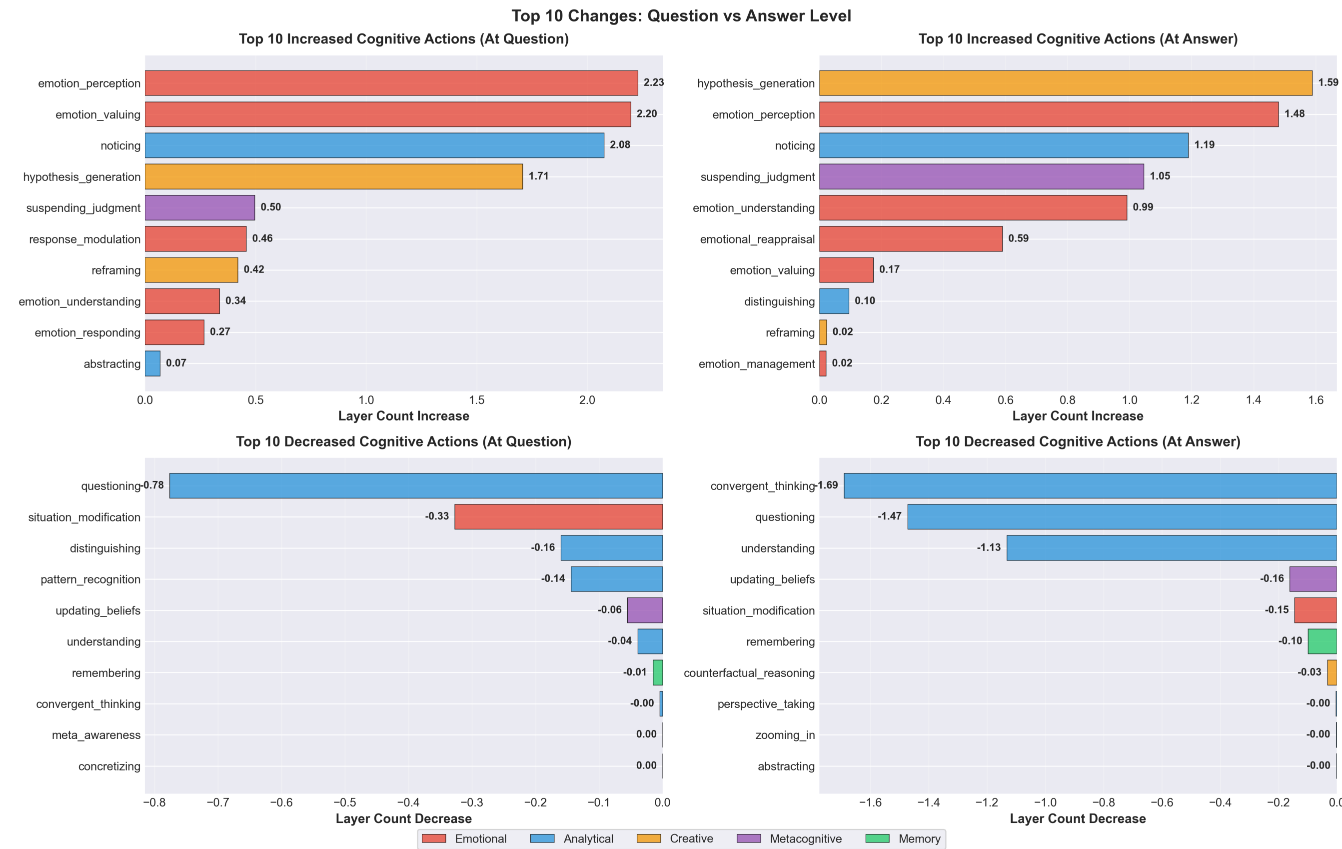

Figure 4: Top 10 cognitive actions with largest increases and decreases between baseline and steered conditions at question level (left) and answer level (right). Emotional actions (emotion_perception, emotion_valuing, noticing) show the strongest upregulation, while analytical actions (questioning, convergent_thinking, understanding) show the strongest downregulation, revealing the cognitive processes most affected by successful ToM steering.

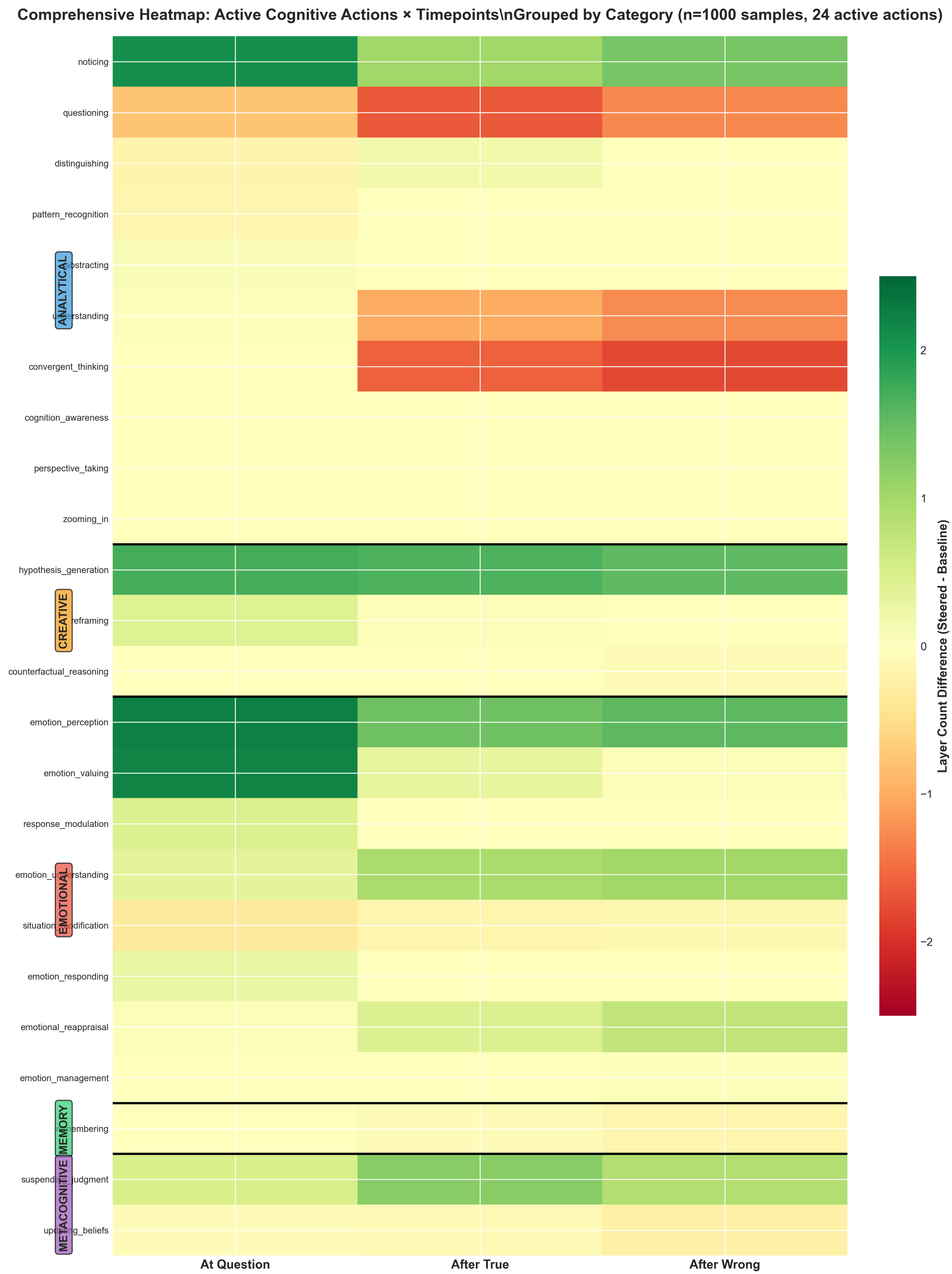

Figure 5: Heatmap of cognitive action activation differences (steered - baseline) across timepoints. Each row represents a cognitive action, and columns show the three measurement timepoints: at question, after true answer, and after wrong answer. Green indicates upregulation and red indicates downregulation.

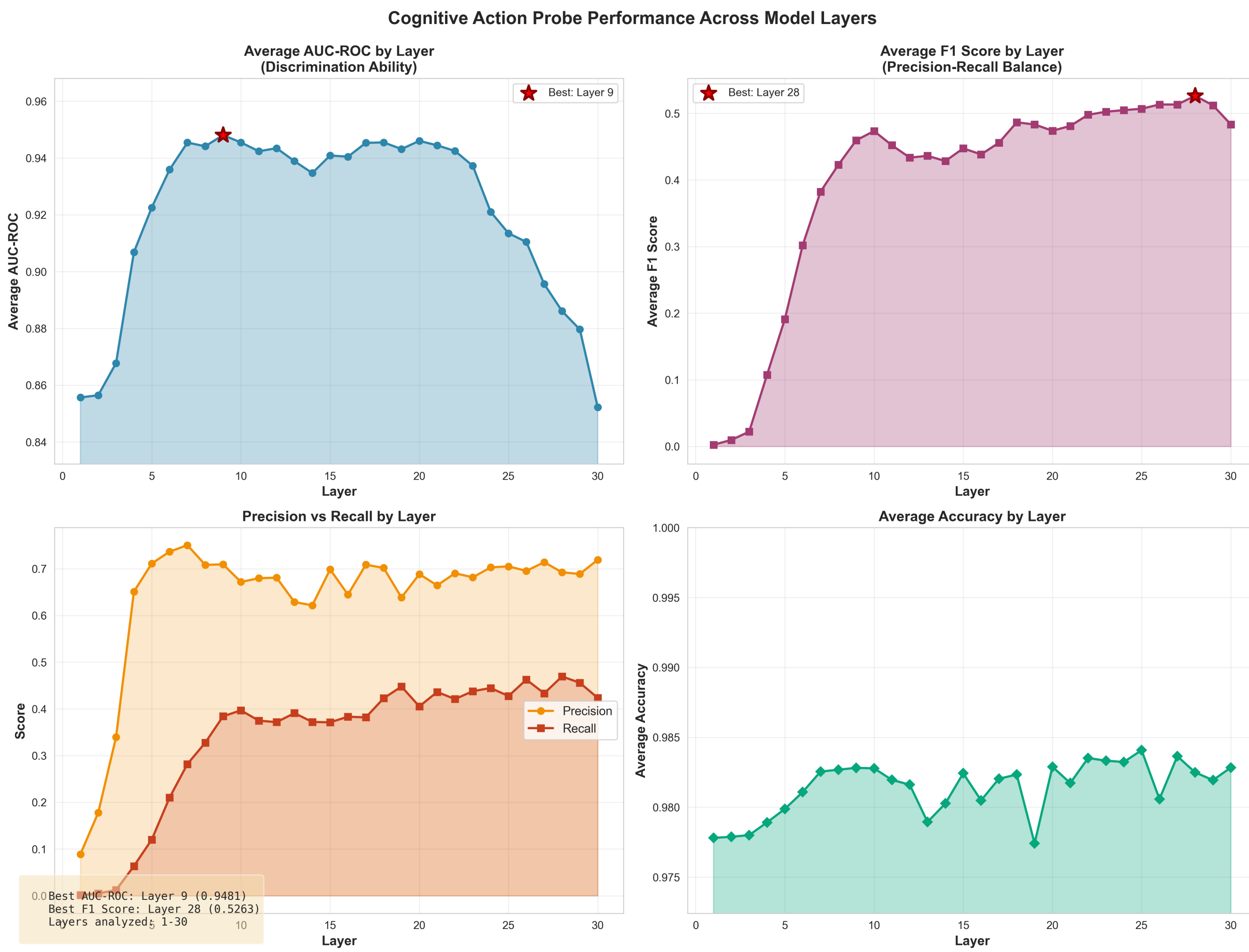

Figure 6: Cognitive action probe performance across all 30 layers of Gemma-3-4B. The visualization shows average AUC-ROC scores for each layer, with Layer 9 achieving peak performance (0.948 AUC-ROC). Strong performance is maintained across mid-layers (5-24), while early and late layers show degraded performance. This pattern suggests that early layers focus on surface-level features, mid-layers capture high-level cognitive abstractions, and late layers optimize for next-token prediction, potentially overwriting intermediate representations.

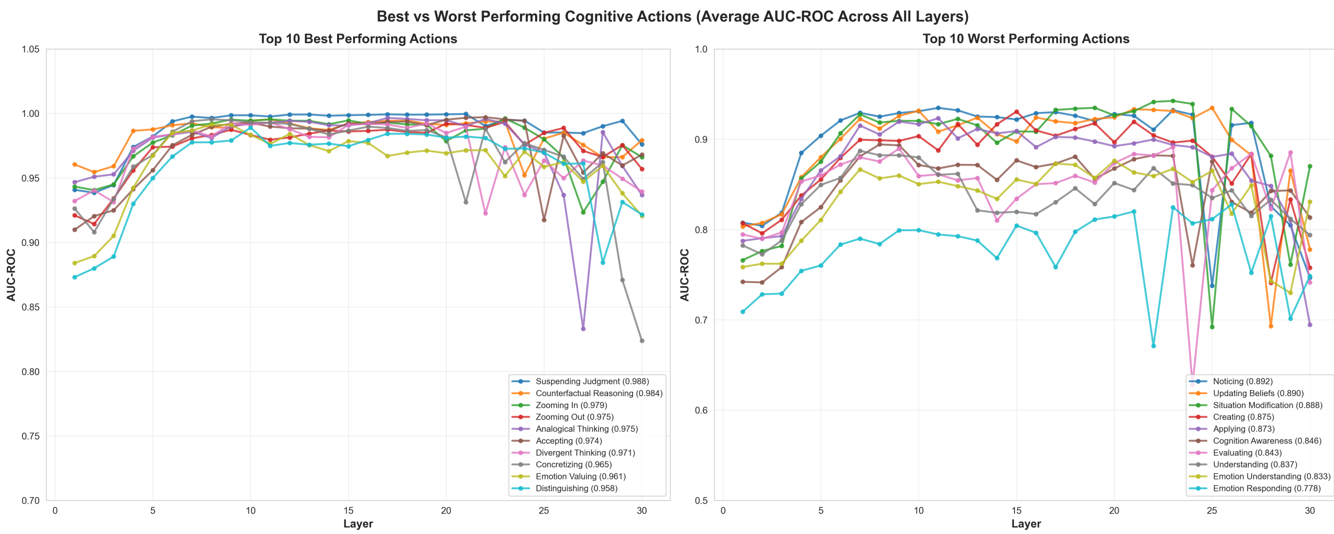

Figure 7: Comparison of top 10 and bottom 10 performing cognitive actions ranked by average AUC-ROC across all layers. Best performers like *suspending_judgment* (0.988) and *counterfactual_reasoning* (0.984) show consistently high performance and distinct activation patterns across most layers. Worst performers like *emotion_responding* (0.778) and *understanding* (0.837) exhibit more variability and lower overall discrimination ability, suggesting these concepts may be more distributed or context-dependent in the model's representation space.